

- Q.26 Write a short note on "Runways used in Airports".
- Q.27 Write a short note on use of road pavers during the construction of highways.
- Q.28 Describe the procedure of repair of pot holes in flexible pavements.
- Q.29 Describe the necessity of providing the pipe drains in hill roads. (Any five)
- Q.30 Write a short note on open dressing with seal coat type of pavement surfacing.
- Q.31 Write any five advantages of construction of highways using flexible pavements.
- Q.32 Describe the process of stabilization of soil in highways using cement.
- Q.33 Write the advantages of use of geo synthetics in construction of highways in hilly areas. (any five)
- Q.34 Write a short note on use of cut back used as binders for road materials.
- Q.35 Describe the remedial measures to be adopted for control of soil erosion during the construction of highways in hilly areas.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain the conduct of laboratory procedure to determine the CBR Value of materials used in construction of highways.
- Q.37 Explain the working process of a hot mix plant used for the construction of highways.
- Q.38 Explain the procedure of construction of highways using the Water Bound Macadam technique.

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5th Sem / Civil, Constr. Mgmt., Civil Engg. (Spl. Highway Engg.) Sub : Highway Engineering

Time : 3Hrs.

M.M. : 100

SECTION-A

- Note:** Multiple choice questions. All questions are compulsory (10x1=10)
- Q.1 Minimum of 8m width and 2m wide shoulders on the following roads are recommended_____
- Express highway and National highway
 - National highway and state highway
 - Express highway, National highway and state highway
 - Major district road
- Q.2 Camber is provided in the road for _____.
 - Counteracting the centrifugal force
 - Effective drainage
 - Getting better sighting distance
 - Getting least stopping distance
- Q.3 The length of vehicle controls the design of _____.
 - Gradient
 - Camber
 - Overtaking distance
 - All the above
- Q.4 Length of the transition curve is decided on the basis of _____.
 - Rate of super elevation
 - Time rate
 - Rate of change of radial acceleration
 - Higher of the above

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- Q.5 Dowel bars are used in rigid pavements _____
a) To distribute pressure on base
b) To take care of temperature stresses
c) To prevent warping
d) All the above
- Q.6 Equivalent passenger car unit in unit time of a truck is _____
a) 0.5
b) 1.0
c) 3.0
d) 6.0
- Q.7 Accident studies are useful for _____
a) Future improvements of roads
b) For improving signal systems
c) For making a road one way
d) All the above
- Q.8 Which one of the following is not a regulatory sign?
a) No U-turn
b) Overtaking prohibited
c) School zone
d) No parking
- Q.9 If more than one runways are to be provided, the basic pattern of runway is _____
a) Parallel
b) Intersecting
c) Non-intersecting
d) Any of the above
- Q.10 Runway markings are to indicate _____
a) Centre line
b) Edge
c) Touchdown zone
d) All the above

SECTION-B

Note: Objective type questions (select the appropriate option).
All questions are compulsory. (10x1=10)

- Q.11 Maximum camber recommended by IRC is _____
(1.7%/4%)

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